

2-Channel Time Interleaved ADC

Team Aevulog

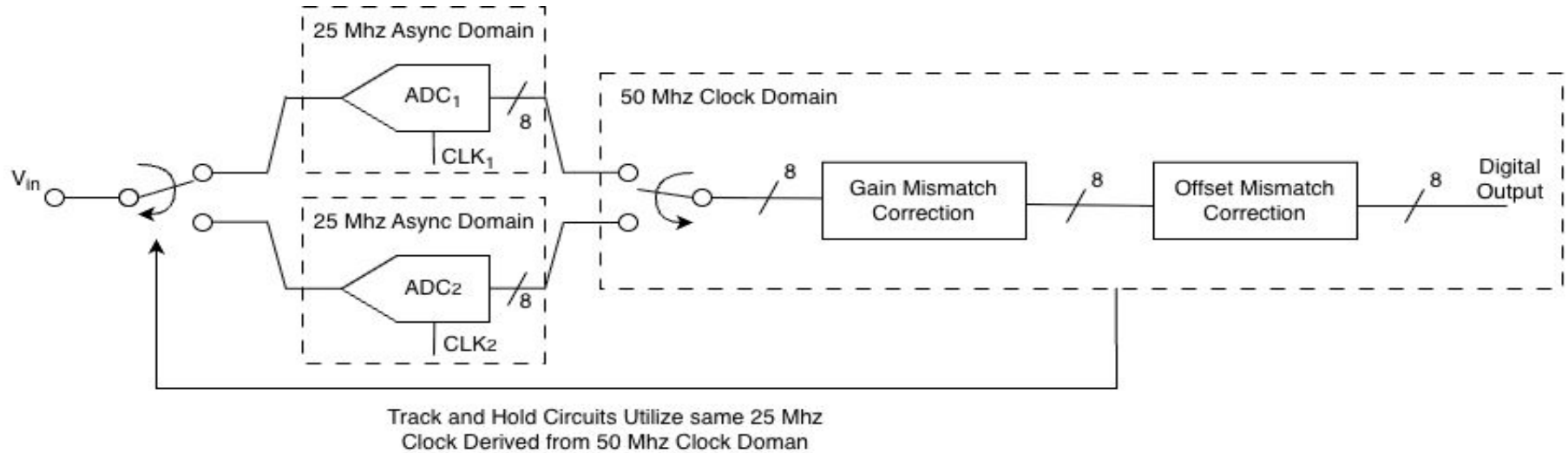
About Team

Name	Discord name	Affiliation (experience)	Role
Shreyas Angadi	Shreyas	Georgia Institute of Technology (Undergrad)	Digital Designer (Team Lead)
Andrés De La Torre	baggy_star	Georgia Institute of Technology (Undergrad)	Analog Designer
Logan Whitehead	Logan Whitehead	Georgia Institute of Technology (Undergrad)	Analog Designer
Lena Conde Araujo	Iena	Olin College of Engineering (Undergrad)	Analog Designer

High Level Overview

- Two 8-bit SAR ADCs are interleaved to achieve 50 MS/s throughput on GF180MCU, each channel sampling at 25 MS/s
- Each SAR ADC uses a double-tail comparator and a split-array CDAC, reducing capacitor count from 256 to 33 for lower power and faster operation
- Digital background calibration corrects offset and gain mismatch between channels without interrupting operation

Annotated block diagram



Gain Mismatch Digital Correction

- Dynamically corrects inter-channel gain mismatch in a time-interleaved architecture by treating CH0 as the reference and scaling CH1 to match its signal envelope.
- Accumulates the absolute values of both ADC channels over a configurable sample window to extract the average signal magnitude and calculate the instantaneous gain error.
- Adjusts a fixed-point gain coefficient continuously using an LMS-style feedback loop, utilizing a computationally cheap, right-shifted step size (μ) instead of complex division.
- Features an activity-monitoring threshold to freeze the calibration loop during DC or low-signal periods, preventing coefficient divergence, alongside a 2-stage pipelined datapath with saturation logic to prevent overflow.

J. Elbornsson, F. Gustafsson and J. -E. Eklund, "Blind adaptive equalization of mismatch errors in a time-interleaved A/D converter system," in *IEEE Transactions on Circuits and Systems I: Regular Papers*, vol. 51, no. 1, pp. 151-158, Jan. 2004

Offset Mismatch Digital Correction

- Compensates for DC offset mismatch between time-interleaved channels by dynamically aligning the baseline of CH1 to the CH0 reference.
- Averages incoming signed samples over a configurable, power-of-two window to filter out AC signal components and isolate the instantaneous DC error.
- Updates the offset correction value using a computationally light, shift-based step size (μ) into a high-precision register, preserving fractional bits to ensure stable, fine-grained convergence.
- Subtracts the rounded, integer-converted offset from CH1 in a pipelined stage, utilizing built-in saturation logic to protect the final output against numerical overflow or wrapping.

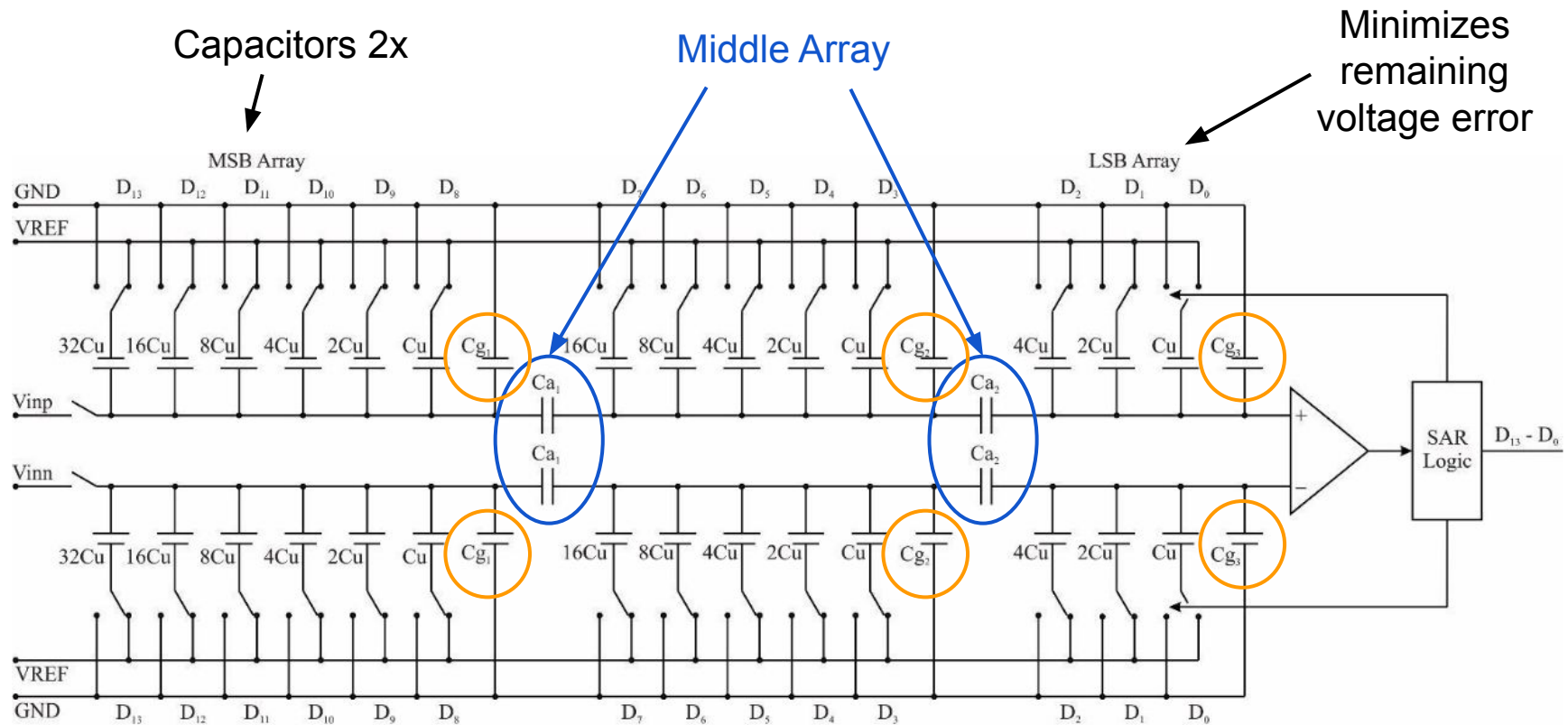
Daihong Fu, K. C. Dyer, S. H. Lewis and P. J. Hurst, "A digital background calibration technique for time-interleaved analog-to-digital converters," in *IEEE Journal of Solid-State Circuits*, vol. 33, no. 12, pp. 1904-1911, Dec. 1998

Split-array CDAC

- 33 capacitors vs. 256 for full-binary weighted, allowing for a 8x area reduction compounding across both channels
 - Dual split (MSB and LSB)
 - Smaller total capacitance → Faster and noise target met
- Switching energy scales as $1/2CV^2$, meaning, smaller array = lower power
- **Main trade-off:** Bridge cap sizing and a more involved switching sequence

Splitarray origin: B. Ginsburg & A. Chandrakasan, "500-MS/s 5-bit ADC in 65-nm CMOS," *JSSC* 2006

Hariprasath et al., "Merged capacitor switching based SAR ADC," *Electronics Letters* 2010



Capacitors 2x

Middle Array

Minimizes remaining voltage error

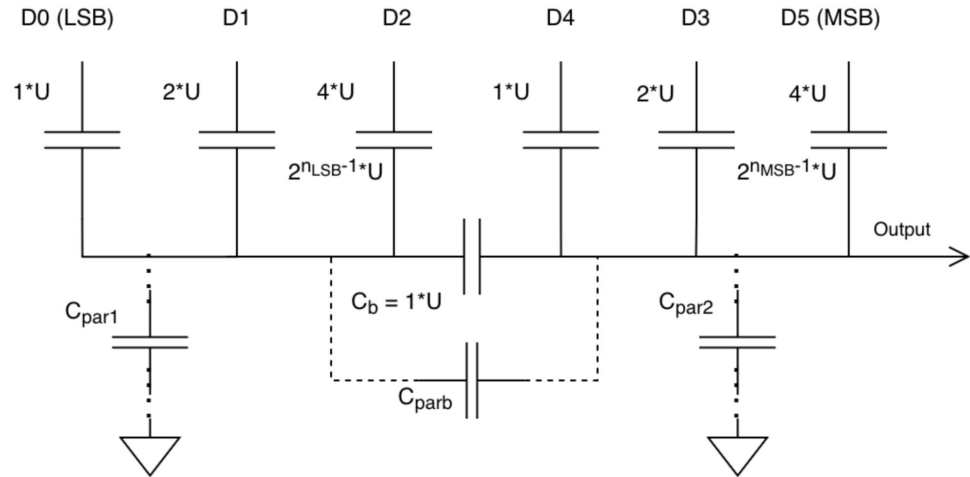
$$V_{REF} = 1V, Cu = 8fF, 2Cu = 16fF, 4Cu = 32fF, 8Cu = 64fF, 16Cu = 128fF, Cg_1 = 256fF, Cg_2 = 248fF, Cg_3 = 392fF, Ca_1 = 16fF$$

C_g : Dummy capacitors
 Ca_1 & Ca_2 : Coupling capacitors

Split-array CDAC Layout

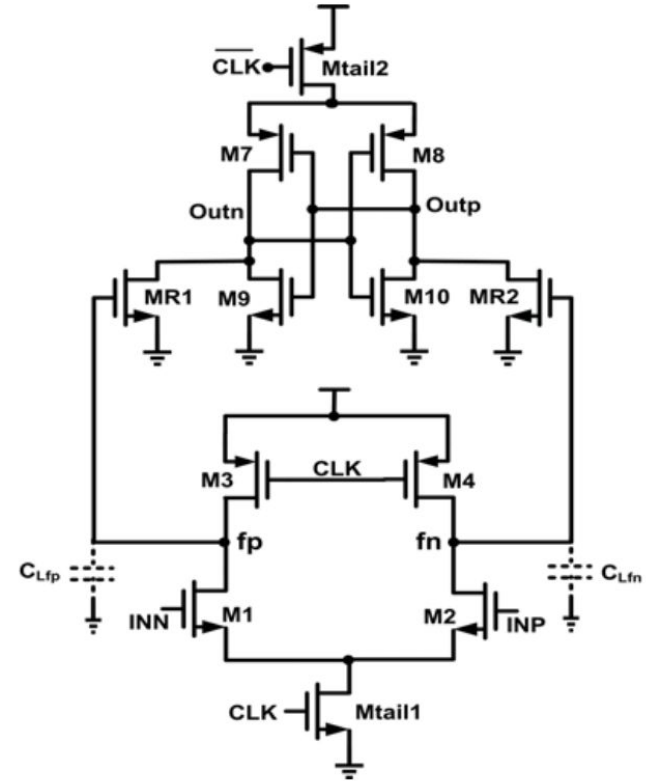
- 8-bit requires $\sim 0.4\%$ matching ($1/2^8$), achievable in GF180MCU without calibration
- Common centroid layout minimizes mismatch; shielding reduces parasitic sensitivity

"Capacitor mismatch effects in SC circuits," /SCAS 1990



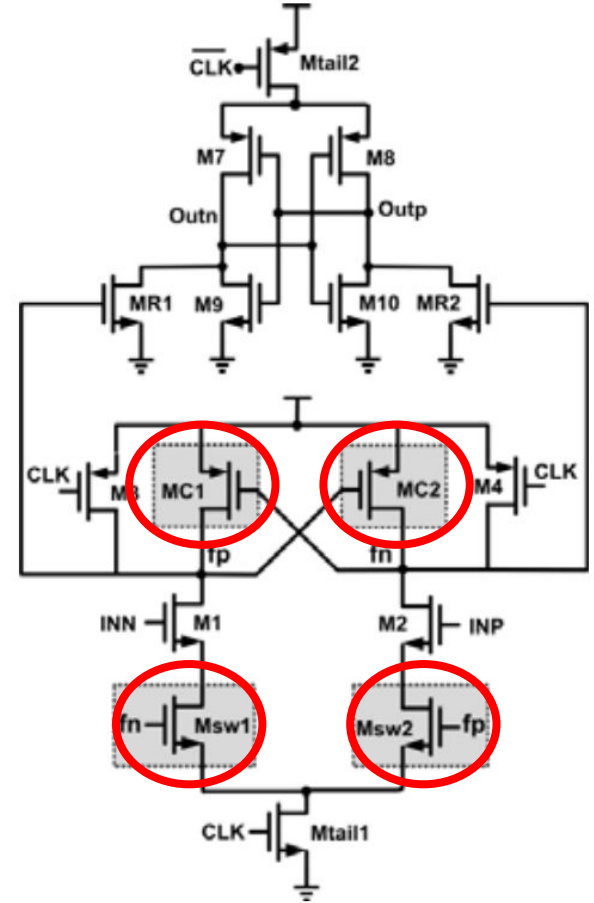
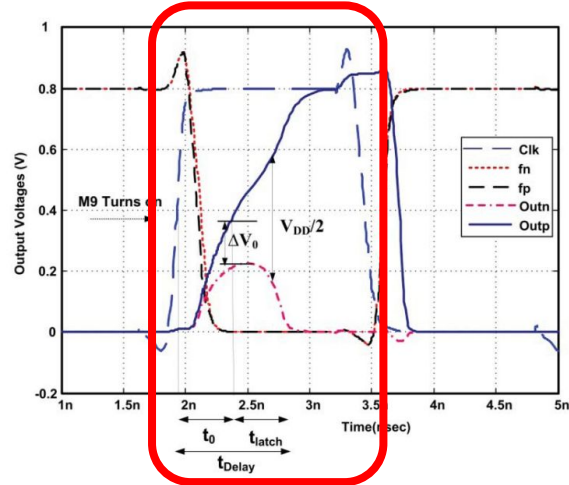
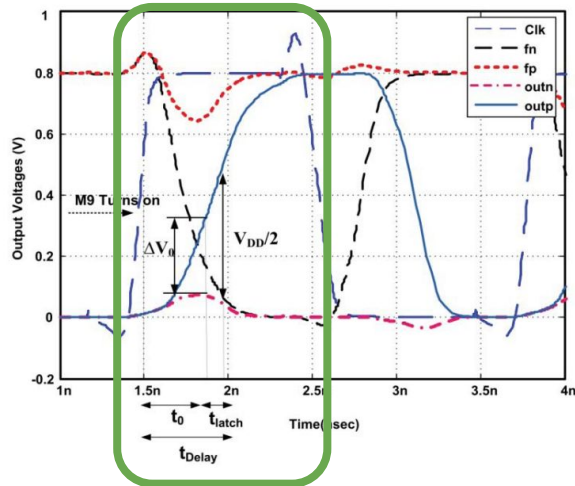
Double-tail comparator

- Optimized speed at 8 bits
- Better offset than StrongARM, still very fast
- At CLK = 1, both tails are open
- M3 and M4 closed
- If $INN > INP$, fp discharges quicker than fn
- MR1 closes faster than MR2
- Outn is higher than Outp, which causes M10 to discharge and amplify difference



Upgraded Double-tail comparator

- Cross coupled transistors (MC1/MC2):
 - Pull the node that discharges faster (either fp or fn) to VDD
 - Greater voltage difference between fn and fp, quicker latch regeneration



Upgraded Double-tail comparator

- Bottom switches (Msw1/Msw2):
 - When MC1 or MC2 are allowing current in, to stop it and avoid static power

